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Youssef Adel

COMPUTER SCIENCE STUDENT | DATA ENGINEERING & AI

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PROFESSIONAL SUMMARY

Motivated Computer Science undergraduate (GPA 3.44 / 4.00) with a solid foundation in Data Engineering, Machine Learning, and Algorithm Design. Experienced in building automated data pipelines, implementing ML algorithms from scratch, and extracting actionable insights from structured and unstructured data. Proficient in Python (Pandas, NumPy, Scikit-Learn), SQL, and C++, with growing expertise in Network Security fundamentals. Proven leadership through national youth tech initiatives presented to government ministries and international organizations. Seeking an AI / Data Engineering internship to apply analytical and development skills in a real-world technical environment.

EDUCATION

October University for Modern Sciences and Arts (MSA University) Cairo, Egypt

Bachelor of Science in Computer Science Expected Graduation: July 2027

Cumulative GPA: **3.44 / 4.00**

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems (SQL), Artificial Intelligence, Machine Learning, Computer Networks, Object-Oriented Programming (C++), Operating Systems, Computer Architecture.

TECHNICAL SKILLS

Languages	Python, SQL, C++, JavaScript
Data & AI Libraries	Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn
Data Engineering	ETL Pipeline Design, Data Cleaning & Transformation, Data Quality, Automated Pipelines
Machine Learning	Classification, Regression, Clustering (KNN, SVM, Linear Regression, K-Means) — implemented from scratch and with Scikit-Learn
Cybersecurity & Networks	IP Addressing, Subnetting, Network Hardware, Threat Analysis Concepts
Tools & Platforms	Git & GitHub, VS Code, Jupyter Notebook, Linux CLI
Soft Skills	Problem-Solving (CPC Competitor), Leadership, Stakeholder Presentations, Analytical Thinking

PROJECTS

Machine Learning Algorithms From Scratch github.com/Youssef-Adel91/ML-Algorithms-From-Scratch

Python · NumPy · Pandas · Scikit-Learn

- Implemented core ML algorithms (KNN, SVM, Linear Regression, K-Means Clustering) in pure Python without high-level ML abstractions, demonstrating deep understanding of mathematical foundations.
- Leveraged NumPy for optimized matrix operations and Pandas for data preprocessing and feature engineering.
- Conducted comparative performance benchmarking against Scikit-Learn implementations to validate accuracy and computational efficiency.

InfoDemics: Data Engineering & Analytics Pipeline
github.com/Youssef-Adel91/InfoDemics

Python · PySpark · SIR Modeling · Data Visualization

- Designed and built an end-to-end data pipeline ingesting, cleaning, and transforming a social network graph of **160+ nodes and 260+ edges** extracted from real Twitter interaction data.
- Applied epidemiological SIR (Susceptible-Infected-Recovered) modeling to simulate misinformation spread across **3 distinct propagation scenarios**, quantifying infection rates and recovery thresholds.
- Produced **10+ analytical visualizations** (heatmaps, spread curves, network graphs) to surface actionable insights on slowing viral falsehoods — translating raw data into business intelligence.

ML Comparative Analysis — Algorithm Benchmarking
github.com/Youssef-Adel91

Python · Scikit-Learn · Pandas · Matplotlib

- Benchmarked **4 classification algorithms** (SVM, Random Forest, XGBoost, KNN) across **3 structured datasets**, evaluating Accuracy, Precision, Recall, and F1-Score.
- Generated confusion matrices and ROC curves to identify the optimal model candidate per dataset, reducing misclassification rate by selecting best-fit hyperparameters.
- Produced a comparative analysis report demonstrating trade-offs between model complexity and predictive performance across **12+ experimental runs**.

Diabetes EDA — Health Data Analytics
github.com/Youssef-Adel91

Python · Pandas · Seaborn · Statistical Analysis

- Conducted deep exploratory data analysis on a healthcare dataset of **768 patient records** and **8 clinical features** to uncover key diabetes risk indicators.
- Identified and treated **5 zero-value anomalies** (physiologically invalid entries) through targeted imputation, improving dataset integrity before modeling.
- Produced **15+ statistical visualizations** (correlation heatmaps, distribution plots, pair plots) demonstrating strong feature relationships and class-specific patterns.

LMS Software Requirements Specification — System Architecture
SQL · UML · System Design

UML · SRS · Use Case Design · Relational Database Modeling

- Authored a **90-page IEEE-830 compliant Software Requirements Specification (SRS)** for a Learning Management System, covering all functional and non-functional requirements.
- Designed **20+ UML diagrams** (Use Case, Sequence, Class, Activity) mapping interactions across **6 system actors** and **30+ use cases**.
- Structured a full relational database schema (ERD, 3NF normalization) with stored procedures and complex SQL queries for reporting and data retrieval.

CERTIFICATIONS

- **IBM Data Science Professional Certificate** — IBM / Coursera
- **IBM Introduction to Data Engineering Certificate** — IBM / Coursera
- **Huawei Certified ICT Associate – Artificial Intelligence (HCIA-AI V4)** — Huawei ICT Academy
- **Problem Solving Training** — CPC (Collegiate Programming Contest) Preparation

VOLUNTEERING & LEADERSHIP

Maharat Misr National Initiative

Oct 2024

Team Leader

- Led a cross-functional team to design and implement a technology-driven solution integrating digital tools with academic learning, with a specific focus on accessibility for people with disabilities.
- Presented project outcomes and technical proposals to high-level stakeholders, including representatives from **Egypt's Ministry of Youth and Sports** and **GIZ (Deutsche Gesellschaft für Internationale Zusammenarbeit)**.

Lebaladna Initiative

March 2025

Team Leader

- Managed team logistics, task delegation, and operational planning to enhance organizational efficiency throughout community charity initiatives.
- Coordinated cross-team communication and ensured timely delivery of project milestones.